

Movement and crossover asymmetries in the Seediq pivot*

Yvette Yi-Chi Wu

Harvard University

1. Introduction

Recent work (Keine and Bhatt 2023, Chierchia 2025) has called to attention a seemingly diverse range of constructions that obviate *weak crossover* (WCO) but not *strong crossover* (SCO). I show that in Seediq—an Austronesian language of Taiwan—promotion to “pivot position” displays similar crossover asymmetries as well, providing novel data on primary and secondary SCO; at the same time, *wh*- and focus elements are generally prohibited in pivot position and rely on other strategies such as *wh*-in-situ or predicate clefting.

These facts provide support for the idea that the *A/Ā distinction* should be reframed in terms of *static vs. dynamic binding* (Chierchia 2020; cf. Büring 2004) and the potential of relevant functional heads—such as C or T—to introduce discourse referents. The result is that the ability of topic and A-movement (as opposed to *wh*- and focus related movement) to create new binding relations can be attributed to the same mechanism. In addition, given cross-linguistic comparisons to English, Italian, and Hindi, some initial speculation can be made about the (un)availability of this mechanism in *wh*-question formation that depends on the employment of overt vs. covert *wh*-movement in the language. Overall, these facts are also in line with previous *Ā*-analyses of Austronesian voice systems, where pivots are topic-like elements (Schachter 1976, Richards 2000, Pearson 2005, Chen 2017).

2. Background

Austronesian voice systems have long been debated for whether they are more aptly characterized as *subject* (A) or *topic* (*Ā*) constructions (Schachter 1976, Guilfoyle et al. 1992, Aldridge 2004, Pearson 2005, Rackowski and Richards 2005, Chen 2017, a.o.)—or a combination of both (composite or “mixed” A/*Ā* accounts, van Urk 2015, a.o.).

*My deepest gratitude to my Seediq language consultants pai Habo, pai Lubi, ptgesa Takun, ptgesa Pihu, Mahung, and ptgesa Tado, as well as Jonathan Bobaljik, Gennaro Chierchia, Natasha Thalluri, and audiences at NELS 55 for their helpful discussion. The Tgdaya Seediq data throughout this paper comes from primary fieldwork conducted in Alang Gluban in Taiwan 2022–2024. Any and all errors are my own.

The main properties of Austronesian voice systems are typically described as follows:

- (1) a. *voice morphology*: morphology on the verb reflects the choice of pivot
- b. *pivot*: each clause has a morphologically/structurally privileged argument
- c. *extraction restriction*: the pivot is the sole argument that can be $\bar{A}$ -extracted

One challenge in this debate is that these *pivots* do not behave straightforwardly in terms of classic subjecthood (Dixon 1979) or A/ $\bar{A}$ diagnostics (Chomsky 1981, Safir 2019). The tables below summarize the behavior of pivots in Seediq (largely based on recent literature that compares these diagnostics across Austronesian; Katochoritis and Lohninger 2024):

TABLE 1: subjecthood diagnostics

control	reflexives	imperatives
n/a	x	x

TABLE 2: A/ $\bar{A}$ diagnostics

reconstruction	WCO	SCO
✓	x	✓

In general for Seediq, pivots fail subjecthood diagnostics such as the licensing of reflexive anaphors (2) or imperative addressees (Holmer 1996); that is, regardless of which argument is promoted to pivot position, the anaphor *heya nanaq* must always be the patient (2a–b). The promotion of a patient to pivot position does not license agent anaphors (2c), as is the case in passive constructions in English where the patient becomes the derived subject.

- (2) a. q⟨m-n⟩ita **heya nanaq** [ka Temi]
 ⟨AV-PFV⟩see 3SG self PIV Temi
 ‘*Temi* saw herself.’ (NOTE: pivots are italicized) actor voice (AV)
- b. q⟨n⟩ita na Temi [ka **heya nanaq**]
 ⟨PV.PFV⟩see GEN Temi PIV 3SG self
 ‘*Temi* saw *herself*.’ patient voice (PV)
- c. *q⟨n⟩ita na **heya nanaq** [ka Temi]
 ⟨PV.PFV⟩see GEN 3SG self PIV Temi

Thus, pivots do not appear to align cross-linguistically with what we know of subjects. At the same time, pivots yield seemingly mixed results with respect to A/ $\bar{A}$ diagnostics. One notable observation is that promotion to pivot position obviates WCO (3) (Chen 2017; see similar facts in Tagalog and Malagasy, Rackowski 2002, a.o.), which is typically seen as a behavior of A-movement. That is, the quantified patient *knkingal laqi* cannot bind into the agent possessor *daha* when remaining in-situ (3a), but can when it is promoted and moves

- (3) a. #wada m-emux **kn~kingal laqi**₂ [ka bubu=**daha**₂]
 PST AV-hug RED~one child PIV mother=3PL.GEN
 ‘*Their*₂ mother hugged every child₂.’ actor voice (AV)
- b. wada mex-un na bubu=**daha**₂ [ka **kn~kingal laqi**₂]
 PST hug-PV GEN mother=3PL.GEN PIV RED~one child
 ‘*Their*₂ mother hugged *every child*₂.’ patient voice (PV)

to the sentence-final pivot position (3b). Interestingly, however, pivots also reconstruct for quantifier-variable binding, which is typically seen as a behavior of $\bar{A}$ -movement.

These conflicting behaviors have led scholars to question how Austronesian voice systems fit into the typology of movement more generally. Meanwhile, in another segment of the literature, an increasing list of (seemingly disparate) constructions have been observed to exhibit similar behaviors (see Table 3) (Keine and Bhatt 2023, Chierchia 2025). These include English PP dislocation, Italian clitic left dislocation, and Hindi local scrambling, which along with Seediq pivot movement would not be considered subject forming constructions, but have been shown to systematically obviate WCO despite corresponding $\bar{A}$ -constructions in the language—like *wh*-movement—not being able to do so.

TABLE 3: constructions cross-linguistically with crossover asymmetry

	WCO (but not SCO) obviated	WCO not obviated
English	PP dislocation	<i>wh</i> -movement, quantifier raising
Italian	clitic left dislocation	<i>wh</i> -movement, quantifier raising
Hindi	local scrambling	<i>wh</i> -in-situ, quantifier raising
Seediq	pivot movement	<i>wh</i> -in-situ, quantifier raising

Additionally, these constructions are observed to exhibit a systematic *asymmetry* between their ability to obviate WCO but not SCO. I show similar data below for SCO in Seediq, both for primary (4) and secondary (5) SCO. In contrast to the WCO examples in (3), the same quantified patient *knkingal seediq* can never bind a variable or anaphor *heya / dheya* (*nanaq*) in agent position, regardless of whether the patient is in-situ (a) or moved to pivot position (b) (see Keine and Bhatt 2023 for discussion of primary vs. secondary crossover).

- (4) a. #wada m-ita **kn~kingal seediq₃** [ka **heya / dheya (nanaq₃)**]
PST AV-see RED~one person PIV 3SG 3PL self
‘They/Themselves₃ saw every person₃.’
- b. #wada qta-un na **heya / dheya (nanaq₃)** [ka **kn-kingal seediq₃**]
PST see-PV GEN 3SG 3PL self PIV RED-one person
‘They/Themselves₃ saw *every person*₃.’
- (5) a. #wada m-angal hndayu na **kn~kingal seediq₃** [ka **heya / dheya₃**]
PST AV-take lunchbox GEN RED-one person PIV 3SG 3PL
‘They₃ took every person₃’s lunchbox.’
- b. #wada ngal-un na **heya / dheya₃** [ka hndayu na **kn~kingal seediq₃**]
PST take-PV GEN 3SG 3PL PIV lunchbox GEN RED-one person
‘They₃ took *every person*₃’s lunchbox.’

This stands in contrast to A-movement in English, which obviates WCO *and* SCO (Postal 1968; see discussion below for a reconsideration of these facts). Note that this asymmetry is unexpected if co-indexation here was considered a matter of coreference—and not binding—as there would not be a difference in behavior between (secondary) WCO and SCO.

The emerging sentiment in the literature, therefore, has been that the $A/\bar{A}$ distinction is perhaps not the most adequate system to characterize the typology of movement. While a line of research has attempted to address this issue through a notion of mixed $A/\bar{A}$ movement and composite probes (van Urk 2015, Colley and Privoznov 2020, a.o.), it is worth considering a need for greater reevaluation in how movement should be categorized, and the primitive syntactic and/or semantic mechanisms that underlie such categories.

3. Account

This account implements a framework of *bimodal binding* (Chierchia 2020, 2025), which proposes that the core binding mechanism utilized in A-movement is also utilized in topic movement. A key consequence is that the notion of “ $\bar{A}$ ” needs to be internally decomposed, such that the divide in binding mechanisms lies instead between topic and *wh*-movement.

Two basic components of this framework are: (i) the division in binding between static vs. dynamic variables, and (ii) the introduction of discourse referents at dedicated syntactic positions. In particular, there are separate channels of binding for traces and pronouns (cf. Buring 2004), where traces are *static variables* bound by lambda extraction λx_n (6a) while pronouns are *dynamic variables* bound across conjuncts of a $^n()$ -operator (6b). The former is employed in phenomena where movement occurs for the purposes of scope assignment (e.g., QR and *wh*-movement), while the latter is employed when movement is accompanied by the introduction of a new discourse referent—in this case subject and topic movement.

- (6) a. static variables: $\lambda x_n [\alpha \dots x_n \dots] \dots x_n \dots$
 b. dynamic variables: $[\beta \ ^n(u) \wedge \dots \omega_n \dots \wedge \dots \omega_n \dots \wedge \dots]$

An important logical principle here is that dynamic binding is correlated with but ultimately *not* determined by c-command, as the scope of a $^n()$ -operator extends across its conjuncts whose order is not transitive and does not necessarily reflect structural hierarchy.

Furthermore, under the *dynamic predication principle* there are dedicated syntactic positions that introduce new *discourse referents* (DRs) into the common ground. While this is typically associated with base-merged thematic positions (e.g., AG and TH), a key assumption here is that this holds true of subject and topic positions (e.g., T and C) as well; thus, this allows both topic and A-movement the possibility of creating new binding relations. A general semantics of a functional head that activates a DR is shown in (7a), where it takes a type $\langle e, t \rangle$ predicate and a type e entity, and asserts that that entity is established as a DR at the index of n and that the predicate holds true of that entity. For concreteness, the head TOP^n takes as its complement a predicate Φ and as its specifier a constituent XP_n (which originates within Φ), and depending on the type of XP_n yields the semantics in (7b). The tree in (8) illustrates the derivation in more detail, with the TOP^n operator in C.

- (7) a. $\llbracket \text{TOP}^n \rrbracket = \lambda P \lambda u. \ ^n(u) \wedge P(u)$
 b. $[\text{XP}_n \text{ TOP}^n \Phi]$
 (i) for XP 's of type $e \Rightarrow \ ^n(\text{XP}_n) \wedge \lambda x_n. \Phi(\text{XP}_n)$
 (ii) for XP 's of type $\langle \langle e, t \rangle, t \rangle \Rightarrow \text{XP}_n(\lambda u. [\ ^n(u) \wedge \lambda x_n. \Phi(u)])$

$$\begin{array}{c}
 (8) \qquad \qquad \qquad \text{CP}^n \\
 \qquad \qquad \qquad \text{}^n(\text{XP}_n) \wedge \lambda x_n. \Phi(\text{XP}_n) \\
 \\
 \text{XP}_n \qquad \qquad \lambda u. \text{}^n(u) \wedge \lambda x_n. \Phi(u) \\
 \\
 \qquad \qquad \qquad \text{C}^n \qquad \qquad \lambda x_n. \Phi \\
 \lambda P \lambda u. \text{}^n(u) \wedge P(u) \qquad \qquad n \quad \text{TP} \\
 \qquad \qquad \qquad \qquad \qquad \qquad \Phi
 \end{array}$$

Turning now to Seediq, I assume a simplified clause structure like that above (abstracting away from language specific details), with a TOP^3 head in C such that the pivot moves from a position internal to TP into the specifier of CP. The derivation for the WCO example in (3b) is shown below (adapted from Chierchia 2025 for Hindi local scrambling):

- (9) a. $[\text{CP} [\text{TP wada mex-un} [\text{DP na bubu}=\mathbf{daha}_3] t_3] [\text{DP ka } ^3(\mathbf{kn-kingal laqi}_3)]]$
 b. $[\mathbf{every\ child}_3] \text{TOP}^3 [[\mathbf{their}_3 \text{ mother}] \text{ hugged } t_3] \Rightarrow$
 (i) $[[\mathbf{every\ child}_3] [\lambda u. [^3(u) \wedge \lambda x_3. [[\mathbf{their}_3 \text{ mother hugged } t_3]](u)]]$
 (ii) $\lambda P. \forall z [\text{child}(z) \rightarrow P(z)] [\lambda u. [^3(u) \wedge \lambda x_3. \text{hug}(\omega_3 \text{'s mother})(x_3)(u)]]$
 (iii) $\lambda P. \forall z [\text{child}(z) \rightarrow P(z)] [\lambda u. [^3(u) \wedge \text{hug}(\omega_3 \text{'s mother})(u)]]$
 (iv) $\forall z [\text{child}(z) \rightarrow ^3(z) \wedge \text{hug}(\omega_3 \text{'s mother})(z)]$

In a step-wise derivation, first, the agent and patient arguments are in their base positions, as the TOP^3 operator in the C head merges with TP and activates the dynamic index $\underline{3}$ via the $^3()$ -operator. The head then searches the structure and does two things: 1) it reaches the dynamic variable *daha* ‘their’ and binds it at the index $\underline{3}$, and 2) it searches and finds the quantified NP *knkingal laqi* ‘every child’ that has the static index “3” and moves it into its specifier. This movement leaves behind a static variable t_3 in its base position that is bound via lambda abstraction; but since static and dynamic variables are bound through separate mechanisms, the λ_3 - and $^3()$ -operators do not interfere with each other. The final result is twofold: a new DR has been activated that can store an entity and bind dynamic variables, and a constituent matching the index of the DR has moved to a higher landing site.

One advantage to this framework is its potential to account for the SCO violations observed in these languages, in the sense of being reduced to *minimality violations* (10) (à la Rizzi 1990). In order to obviate SCO in examples such as (4b) or (5b), the TOP^3 head must

$$\begin{array}{c}
 (10) \qquad \qquad \qquad \text{CP} \\
 \\
 \text{C} \qquad \qquad \text{TP} \\
 \text{TOP}^3 \\
 \qquad \qquad [\text{they}_{\underline{3}}] \\
 \qquad \qquad \dots \quad [\text{every person}_3]
 \end{array}$$

bind the agent variable while also moving the quantified patient to its specifier. However, once TOP³ binds the agent, it serves as an intervener between TOP³ and the patient, since both arguments now share the same index 3 and the agent is structurally closer as a target. Crucially, this issue does not arise in WCO because the variable is a possessor embedded *within* the agent—which has a different index—and does not serve as an intervener.

Although this analysis treats binding in topic and A-movement as the result of the same dynamic mechanism, some discussion is needed in terms of their differences with respect to SCO, reflexive binding, and reconstruction. First, the minimality account of SCO above predicts that A-movement should also exhibit crossover asymmetries, although traditional understanding of English is that A-movement obviates SCO (Postal 1968). A preliminary explanation is to posit that the English cases do not violate minimality because the bound variable is embedded with a PP (e.g., [*Every boy*₂'s mother] seems [*to him*₂]₃ to be smart.) which has a distinct index and does not intervene—much like in WCO (see Chierchia 2025 for more discussion). Second, while A-movement in English can feed anaphor binding, we have seen that pivot movement in Seediq does not (2) (see McGinnis 2004 for similar facts in scrambling languages). This is likely due to the nature of reflexives as *subject-oriented* operations distinct from general binding mechanisms, whose main semantic function is to identify an agent with a patient and is restricted to the argument structure domain (Labelle 2008, cf. Bruening 2006 for reciprocals). Thus a derived subject can feed anaphor binding in English because that subject binds into an internal argument (e.g., [*Every child*₂] seems [*to herself*₂] to be smart.), but otherwise a derived *topic* cannot feed anaphor binding of a *subject*, as a general rule for topic movement in the constructions discussed here. Third, it remains to be seen whether similar reasoning can be extended to differences in quantifier-variable reconstruction in topic and A-movement, which is available in the former but not the latter. For instance, there may be structural differences between subjects and topics that affect the option of interpreting a dynamic variable in its base or moved position.

Overall, this account merges the mechanisms that give rise to binding and crossover in topic and A-movement, under the framework of bimodal (static vs. dynamic) binding. The remaining dissimilarities between topic and A-movement, though related to binding, must be explained instead by featural or structural differences independent of general binding.

4. Discussion

This leaves us with a potential new typology of movement that is further decomposed from the canonical binary A/ $\bar{A}$ distinction. In particular, there is a *ternary distinction* (Table 4) between subjects, topics, and *wh*-movement, where topics occupy a sort of middle ground between the other two. The notions of A and $\bar{A}$ here, therefore, do not refer to a dichotomy in behavior with respect to binding, but rather more descriptively to argument structure (A) vs. information structure ($\bar{A}$) types of movement—which have independent importance.

TABLE 4: a typology of A/ $\bar{A}$ -movement

	subject (A)	topic ($\bar{A}$)	<i>wh</i> /focus ($\bar{A}$)
control, reflexives, imperatives	✓	x	x
(obviates) weak crossover	✓	✓	x

On the one hand, as discussed above, there is something special and subject-oriented about the licensing of control, reflexives, and imperatives which is not available to topic and *wh*-constructions. On the other hand, subjects and topics—to the exclusion of *wh*-movement—share the ability to activate novel discourse referents through dynamic binding, a behavior not unreasonable for either phenomena, given their pragmatic function of organizing discourse around a chosen referent (Reinhart 1981, Portner and Yabushita 1998). In contrast, *wh*-movement, focus movement, and QR involve static binding that does not make new DRs, but have the function of altering scope assignment and triggering alternatives.

As far as the bimodal approach to binding, it should be noted that the idea that different binding mechanisms are at play in the $A/\bar{A}$ distinction is not new (cf. Sauerland 2004, Ruys 2000, 2004). However, these accounts were still aligned with the traditional A vs. $\bar{A}$ division, where topic and *wh*-movement were subsumed under the latter category. To maintain this divide, past literature on topicalization and its ability to create novel binding relations has instead explained such behavior as one of *coreference* (cf. *weakest crossover*, Lasnik and Stowell 1991). While this approach has been adequate for well-studied languages like English, where regular topicalization is fairly restrictive and does not permit quantifiers, what recent literature shows is that within a greater cross-linguistic range of topic constructions these generalizations do not hold true; topicalized quantifiers *do* exist and can in fact bind variables. Thus, these accounts still need to redraw the lines in terms of which categories of movement are able to partake in particular binding mechanisms.

One interesting question to ask is what restrains the constituents that can be topicalized across different languages or constructions. The table below illustrates a potential range of variation, where Hindi and Seediq appear to be more permissive than English or Italian:

TABLE 5: possible topicalized constituents across languages

	definite	indefinite	variable	quantified DP	anaphor
English dislocation (*= PP ok)	✓	x*	x*	x*	x*
Italian clitic left dislocation	✓	✓	✓	x	x
Hindi local scrambling	✓	✓	✓	✓	✓
Seediq pivot movement	✓	✓	✓	✓	✓

While there is nothing within the current framework that would account for such variation, there is an intuition here about Hindi scrambling and Seediq pivots being less marked and often obligatory in these languages, thus needing to allow for a greater range of topics.

Aside from the facts discussed here, there is also other support for the idea that pivots should be treated as topics (Schachter 1976, Richards 2000, Pearson 2005, Chen 2017, a.o.) based on their syntactic distribution, semantic interpretation, and discourse functions. One such example is that pivots must either be *definite* or *specific* (Himmelman 1991). This is straightforwardly captured under the current framework, since pivots are referential and also move to take wide scope, both classic properties of topic phrases.

Finally, the last issue to explore is why *wh*- or focus movement does not have access to a dynamic binding mechanism. One potentially straightforward answer is that the core semantic function of these phenomena—whether that is question formation or the triggering of alternatives—is clearly *not* about arranging a discourse around a designated referent. It

thus makes sense that languages would employ separate syntactic strategies to achieve such dissimilar semantic goals. For instance in Seediq, typical *wh*-question formation involves *wh*-in-situ (11a) or predicate clefting (11b), while *wh*-words are in fact *prohibited* in pivot position (11c); this illustrates even a certain incompatibility between the two phenomena. Note that this distribution also applies to arguments with focus elements like *wano* ‘only’.

- (11) a. wada qbhang-un **ima** [ka sisil]?
 PST hear-PV who PIV sisil
 ‘Who heard *the sisil*?’ [wh-in-situ]
- b. **ima** [ka wada m-bahang sisil]?
 who PIV PST AV-hear sisil
 ‘*The one that heard the sisil* is who?’ [predicate cleft]
- c. #wada m-bahang sisil [ka **ima**]?
 PST AV-hear sisil PIV who
 ‘Who heard the sisil?’ (OK: echo question reading) [wh-pivot]

Interestingly, however, *wh*-words are *sometimes* permissible as pivots in the narrow context of echo questions. Furthermore, there is a contrast between the regular question formation strategies (12a–b) and echo questions (12c) in the ability of the *wh*-phrase to obviate WCO.

- (12) a. #g⟨m⟩aalu **ima**₂ [ka bubu=**na**₂]?
 ⟨AV⟩care who PIV mother=3SG.GEN
 ‘*Their*₂ mother cares for who₂?’ [wh-in-situ]
- b. #**ima**₂ [ka gluw-un bubu=**na**₂]?
 who PIV care-PV mother=3SG.GEN
 ‘*The one that their*₂ mother cares for is who₂?’ [predicate cleft]
- c. #gluw-un bubu=**na**₂ [ka **ima**₂]?
 care-PV mother=3SG.GEN PIV who
 ‘*Their*₂ mother cares for *who*₂?’ (OK: echo question reading) [wh-pivot]

We can deduce from this pattern that it is pivot movement and *not* (covert) *wh*-movement that contributes the possibility of binding in (12c), affirming the generalization in Table 4.

In terms of under what conditions topic and *wh*/focus could reasonably target the same argument, a preliminary observation is that it appears to be related to the permissiveness of topicalization in a given language, as well as whether *wh*-movement is *overt* or *covert*. For languages like English and Italian, where topicalization and *wh*-movement are both overt, a competition arises where both types of movement are perhaps syntactically incompatible and/or compete for the same landing site (cf. Pearson 2005). Given the view of *wh*-phrases as *existential quantifiers* (Chierchia 2013), this is consistent with the fact that quantifiers in these languages cannot be topicalized, as seen in Table 5. In comparison, for languages like Hindi and Seediq, where *wh*-movement is covert and topicalized quantifiers are permitted, there is suddenly a possibility of combining the two phenomena. When this occurs it yields

the semantics not of a standard question but an *echo question*, as shown above for Seediq (and appears to be the case for Hindi as well, Thalluri p.c.). This pattern is summarized in Table 6 below. Finally, note that these languages also have separate strategies in conveying *contrastive topics*, which warrant their own investigation in terms of binding behavior.

TABLE 6: overt movement and topic/*wh*-constructions

	topic	overt?	<i>wh</i>	overt?	both
English	PP dislocation	Y	<i>wh</i> -movement	Y	n/a
Italian	clitic left dislocation	Y	<i>wh</i> -movement	Y	n/a
Hindi	local scrambling	Y	<i>wh</i> -in-situ	N	echo question (?)
Sediq	pivot movement	Y	<i>wh</i> -in-situ	N	echo question (?)

5. Conclusion

I have shown that Seediq pivot movement parallels observations of crossover asymmetries in recent literature. These facts provide support for the notion that topic and A-movement utilize the same dynamic binding mechanism to form new binding relations. I also suggest a typology of movement with a three-way contrast between A-movement, topic, and *wh*-/focus movement, where there are some details with respect to SCO, anaphor binding, and reconstruction that need separate explanation. Overall, these facts are in line with previous analyses of Austronesian voice systems. Further cross-linguistic investigation is needed to determine whether the current framework can account for the extent of variation found in these A/ $\bar{A}$ diagnostics across Austronesian languages (and languages in general), and also account for the range in possible constituents allowed in topic related movement.

References

- Aldridge, Edith. 2004. Ergativity and word order in Austronesian languages. Doctoral dissertation, Cornell University.
- Bruening, Benjamin. 2006. The morphosyntax and semantics of verbal reciprocals. Ms., University of Delaware.
- Büring, Daniel. 2004. Crossover situations. *Natural Language Semantics* 12:23–62.
- Chen, Victoria. 2017. A reexamination of the Philippine-type voice system and its implications for Austronesian primary-level subgrouping. Doctoral dissertation, University of Hawai‘i at Mānoa.
- Chierchia, Gennaro. 2020. Origins of weak crossover: When dynamic semantics meets event semantics. *Natural Language Semantics* 28:23–76.
- Chierchia, Gennaro. 2025. Movement and crossover in three languages. *Natural Language & Linguistic Theory*.
- Chomsky, Noam. 1981. *Lectures on government and binding*. Foris.
- Colley, Justin, and Dmitry Privoznov. 2020. On the topic of subjects: composite probes in Khanty. In *Proceedings of NELS 50*. Cambridge, MA: MIT Press.
- Dixon, Robert M.W. 1979. Ergativity. *Language* 55:59–138.

- Guilfoyle, Eithne, Henrietta Hung, and Lisa Travis. 1992. Spec of IP and Spec of VP: Two subjects in Austronesian languages. *Natural Language & Linguistic Theory* 10:375–414.
- Himmelman, Nikolaus P. 1991. The Philippine challenge to Universal Grammar. *Institute for Linguistics, University of Köln Working Papers* 15.
- Holmer, Arthur J. 1996. *A parametric grammar of Seediq*. Lund University Press.
- Katochoritis, Yiannis, and Magdalena Lohninger. 2024. Between topics and subjects: an A/Ā typology of Austronesian pivots. In *Proceedings of WCCFL 42*. UC Berkeley.
- Keine, Stefan, and Rajesh Bhatt. 2023. Crossover asymmetries. Ms., University of California, Los Angeles and University of Massachusetts Amherst.
- Labelle, Marie. 2008. The French reflexive and reciprocal *se*. *Natural Language & Linguistic Theory* 26:833–876.
- Lasnik, Howard, and Tim Stowell. 1991. Weakest crossover. *Linguistic Inquiry* 22:687–720.
- McGinnis, Martha. 2004. Lethal ambiguity. *Linguistic Inquiry* 35:47–95.
- Pearson, Matthew. 2005. The Malagasy subject/topic as an Ā-element. *Natural Language & Linguistic Theory* 23:381–457.
- Portner, Paul, and Katsuhiko Yabushita. 1998. The semantics and pragmatics of topic phrases. *Linguistics and Philosophy* 21:117–157.
- Postal, Paul M. 1968. *Cross-over phenomena: A study in the grammar of coreference*. Holt, Rinehart and Winston.
- Rackowski, Andrea. 2002. The structure of Tagalog: Specificity, voice, and the distribution of arguments. Doctoral dissertation, Massachusetts Institute of Technology.
- Rackowski, Andrea, and Norvin Richards. 2005. Phase edge and extraction: A Tagalog case study. *Linguistic Inquiry* 36:565–599.
- Reinhart, Tanya. 1981. Pragmatics and linguistics: An analysis of sentence topics. *Philosophica* 27.
- Richards, Norvin. 2000. Another look at Tagalog subjects. In *Formal Issues in Austronesian Linguistics*, ed. by Ileana Paul, Vivianne Phillips, and Lisa Travis, 105–116. Kluwer.
- Rizzi, Luigi. 1990. *Relativized minimality*. MIT Press.
- Ruys, E. G. 2000. Weak crossover as a scope phenomenon. *Linguistic Inquiry* 31:513–539.
- Ruys, Eddy G. 2004. A note on weakest crossover. *Linguistic Inquiry* 35:124–140.
- Safir, Ken. 2019. The A/Ā distinction as an epiphenomenon. *Linguistic Inquiry* 50:285–336.
- Sauerland, Uli. 2004. The interpretation of traces. *Natural Language Semantics* 12:63–127.
- Schachter, Paul. 1976. The subject in Philippine languages: Topic, actor, actor-topic, or none of the above. In *Subject and Topic*, ed. by Charles N. Li, 491–518. Academic Press.
- van Urk, Coppe. 2015. A uniform syntax for phrasal movement: a case study of Dinka Bor. Doctoral dissertation, Massachusetts Institute of Technology.